

Seminar 1. Bayesian Reasoning: Solutions

Bayesian Methods

1. A medical test finds a serious disease in a patient. The test is 99% accurate: if the patient has the disease, the test is positive with probability 99%; if the patient does not have the disease, the test is negative with probability 99%. The disease is rare: only one person in 10 000 has it. Find the probability that the patient really has the disease.

Interpretation. Let $D \in \{0, 1\}$ show whether the patient has the disease ($D = 1$). Let $T \in \{0, 1\}$ be the test result ($T = 1$ is a positive result). The problem gives three values:

$$p(D = 1) = 10^{-4}, \quad p(T = 1 \mid D = 1) = 0.99, \quad p(T = 0 \mid D = 0) = 0.99.$$

Find $p(D = 1 \mid T = 1)$.

Idea. The disease causes the test result, so the causal graph is $D \rightarrow T$. The graph gives the joint distribution as a product of two factors:

$$p(D, T) = p(D) p(T \mid D).$$

The problem gives both factors: the distribution of the source node D and the conditional distribution on the edge $D \rightarrow T$. Thus the joint distribution is known. From the joint distribution, you can get any other probability with two operations:

- A marginal probability is the sum of the joint probability over the other variables. For a continuous variable, the sum is an integral.
- A conditional probability is a joint probability divided by a marginal probability.

These two operations together give Bayes' theorem.

Solution.

Step 1. Find the joint probabilities of a positive result.

The probability of a false positive result is $p(T = 1 \mid D = 0) = 1 - 0.99 = 0.01$. Thus

$$\begin{aligned} p(D = 1, T = 1) &= p(D = 1) p(T = 1 \mid D = 1) = 10^{-4} \cdot 0.99 = 0.000099, \\ p(D = 0, T = 1) &= p(D = 0) p(T = 1 \mid D = 0) = 0.9999 \cdot 0.01 = 0.009999. \end{aligned}$$

Step 2. Find the marginal probability of a positive result.

Sum the joint probabilities over D :

$$p(T = 1) = 0.000099 + 0.009999 = 0.010098.$$

Step 3. Divide the joint probability by the marginal probability.

$$p(D = 1 \mid T = 1) = \frac{p(D = 1, T = 1)}{p(T = 1)} = \frac{0.000099}{0.010098} = \frac{1}{102} \approx 0.0098.$$

The patient has the disease with a probability of approximately 1%.

Conclusion. A positive result increases the probability of the disease from 0.01% to approximately 1%. But the patient is still healthy with a probability of 99%. The cause is the low prior probability. In 10 000 people, the test finds approximately 1 sick person and 100 healthy people. For this reason, most positive results of a mass test for a rare disease are false. A second independent test gives more information: if it is also positive, the probability of the disease increases to 0.495.

2. Consider the following probabilistic model. At a seminar, a student answers badly ($A = 1$) or well ($A = 0$). This depends on whether the student has depression ($D = 1$ or $D = 0$) and whether the student went to a party the night before ($P = 1$ or $P = 0$). The party can also give the student a headache ($H = 1$). A bad answer makes the teacher unhappy ($T = 1$). Figure 1 shows the causal relations in the model. The probabilities are:

$p(A = 1 \mid P, D) \mid P \quad D$	$p(H = 1 \mid P) \mid P$	$p(T = 1 \mid A) \mid A$
0.999	0.9	0.95
0.9	0.2	0.5
0.9		
0.01		

$$p(P = 1) = 0.2, \quad p(D = 1) = 0.4.$$

Find $p(P = 1 \mid H = 1)$, $p(P = 1 \mid T = 1)$, and $p(P = 1 \mid T = 1, H = 1)$.

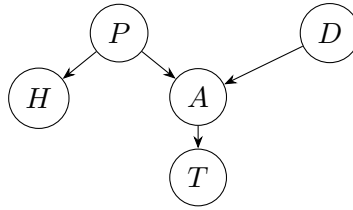


Figure 1: Causal relations for Problem 2

Idea. This problem uses the method of Problem 1 on a larger graph. The graph in Figure 1 gives the joint distribution as a product of one factor for each node. Each factor is the distribution of the node given its parents:

$$p(P, D, A, H, T) = p(P) p(D) p(A \mid P, D) p(H \mid P) p(T \mid A).$$

The tables give all five factors. Thus the joint distribution is known, and you can get any probability from it. To find the probability of P given an observation, move through the graph in two directions:

- Along the edges, from P to the observed variable. Sum over each unobserved variable on the path with the chain rule, for example $p(T \mid P) = \sum_A p(T \mid A) p(A \mid P)$.
- Against the edges, from the observed variable back to P . Use Bayes' theorem, for example $p(P \mid T) \propto p(P) p(T \mid P)$.

Solution.

Step 1. Find the probability of a bad answer given the party.

The depression D is not observed, so sum over D . The variables P and D are source nodes of the graph, so they are independent, and $p(D \mid P) = p(D)$:

$$p(A = 1 \mid P) = \sum_D p(A = 1 \mid P, D) p(D).$$

Substitute the values from the tables:

$$\begin{aligned} p(A = 1 \mid P = 1) &= 0.999 \cdot 0.4 + 0.9 \cdot 0.6 = 0.9396, \\ p(A = 1 \mid P = 0) &= 0.9 \cdot 0.4 + 0.01 \cdot 0.6 = 0.366. \end{aligned}$$

Step 2. Find the probability of an unhappy teacher given the party.

The variable T depends on P only through A , so sum over A :

$$p(T = 1 | P) = p(T = 1 | A = 1)p(A = 1 | P) + p(T = 1 | A = 0)p(A = 0 | P).$$

Substitute the values from Step 1:

$$\begin{aligned} p(T = 1 | P = 1) &= 0.95 \cdot 0.9396 + 0.5 \cdot 0.0604 = 0.92282, \\ p(T = 1 | P = 0) &= 0.95 \cdot 0.366 + 0.5 \cdot 0.634 = 0.6647. \end{aligned}$$

Step 3. Find $p(P = 1 | H = 1)$.

Use Bayes' theorem with the table for $p(H | P)$:

$$p(P = 1 | H = 1) = \frac{p(P = 1)p(H = 1 | P = 1)}{\sum_P p(P)p(H = 1 | P)} = \frac{0.2 \cdot 0.9}{0.2 \cdot 0.9 + 0.8 \cdot 0.2} = \frac{0.18}{0.34} \approx 0.529.$$

Step 4. Find $p(P = 1 | T = 1)$.

Use Bayes' theorem with the values from Step 2:

$$p(P = 1 | T = 1) = \frac{0.2 \cdot 0.92282}{0.2 \cdot 0.92282 + 0.8 \cdot 0.6647} = \frac{0.184564}{0.716324} \approx 0.258.$$

Step 5. Show that H and T are independent given P .

Sum the joint distribution over the unobserved variables D and A . Move each factor out of the sums that do not contain its variables:

$$p(P, H, T) = p(P)p(H | P) \sum_A p(T | A) \sum_D p(D)p(A | P, D) = p(P)p(H | P)p(T | P).$$

The last equality uses Steps 1 and 2. Thus $p(H, T | P) = p(H | P)p(T | P)$.

Step 6. Find $p(P = 1 | T = 1, H = 1)$.

Use Bayes' theorem with the product from Step 5:

$$p(P = 1 | T = 1, H = 1) = \frac{0.2 \cdot 0.9 \cdot 0.92282}{0.2 \cdot 0.9 \cdot 0.92282 + 0.8 \cdot 0.2 \cdot 0.6647} = \frac{0.16611}{0.27246} \approx 0.610.$$

Conclusion. The prior probability of the party is 0.2. A headache increases it to 0.529. An unhappy teacher increases it only to 0.258, because a bad answer has other causes, and the teacher is often unhappy without a bad answer. The two observations together give the highest probability, 0.610.

3. Let $x_1 \sim \text{Poiss}(\lambda_1)$ and $x_2 \sim \text{Poiss}(\lambda_2)$ be two independent Poisson random variables, that is, $p(x_i = k) = \exp(-\lambda_i) \frac{\lambda_i^k}{k!}$, $k = 0, 1, 2, \dots$. Prove that $x_1 + x_2 \sim \text{Poiss}(\lambda_1 + \lambda_2)$.

Idea. To find the distribution of a function $y = f(x_1, \dots, x_n)$ of discrete random variables, find $p(y = k)$ for each value k :

- Write the event $\{f(x_1, \dots, x_n) = k\}$ as a union of disjoint events $\{x_1 = j_1, \dots, x_n = j_n\}$, one for each tuple with $f(j_1, \dots, j_n) = k$.
- Sum the probabilities of these events:

$$p(y = k) = \sum_{f(j_1, \dots, j_n) = k} p(x_1 = j_1, \dots, x_n = j_n).$$

- (c) If the variables are independent, write each joint probability as a product.
- (d) Simplify the sum and compare it with known distributions.

Solution.

Step 1. Write the event as a union of disjoint events.

The variables take only values $0, 1, 2, \dots$. Thus $x_1 + x_2 = k$ if and only if $x_1 = j$ and $x_2 = k - j$ for one $j \in \{0, \dots, k\}$:

$$p(x_1 + x_2 = k) = \sum_{j=0}^k p(x_1 = j, x_2 = k - j).$$

Step 2. Use the independence of x_1 and x_2 .

$$p(x_1 + x_2 = k) = \sum_{j=0}^k p(x_1 = j) p(x_2 = k - j) = \sum_{j=0}^k e^{-\lambda_1} \frac{\lambda_1^j}{j!} e^{-\lambda_2} \frac{\lambda_2^{k-j}}{(k-j)!}.$$

Step 3. Move the factors that do not depend on j out of the sum.

$$p(x_1 + x_2 = k) = e^{-(\lambda_1 + \lambda_2)} \sum_{j=0}^k \frac{\lambda_1^j \lambda_2^{k-j}}{j! (k-j)!}.$$

Step 4. Use the binomial theorem.

Multiply and divide by $k!$. Then the sum is the binomial expansion of $(\lambda_1 + \lambda_2)^k$:

$$p(x_1 + x_2 = k) = \frac{e^{-(\lambda_1 + \lambda_2)}}{k!} \sum_{j=0}^k \binom{k}{j} \lambda_1^j \lambda_2^{k-j} = e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^k}{k!}.$$

This is the probability mass function of $\text{Poiss}(\lambda_1 + \lambda_2)$.

Conclusion. The sum of two independent Poisson random variables is a Poisson random variable, and the parameters add. By induction, the same is true for any finite number of independent Poisson random variables.

4. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. Find the maximum likelihood estimate of λ .

Idea. Use the standard method for a maximum likelihood estimate:

- (a) Write the likelihood: the probability of the sample as a function of the parameter. For an independent sample, it is a product.
- (b) Take the logarithm. The logarithm is an increasing function, so the maximum point does not change. The product becomes a sum.
- (c) Set the derivative to zero and solve for the parameter.
- (d) Make sure that the solution is a maximum, for example with the sign of the second derivative.

Solution. Let $S = \sum_{i=1}^N x_i$.

Step 1. Write the likelihood.

$$L(\lambda) = \prod_{i=1}^N e^{-\lambda} \frac{\lambda^{x_i}}{x_i!}.$$

Step 2. Take the logarithm.

$$\ell(\lambda) = \log L(\lambda) = -N\lambda + S \log \lambda - \sum_{i=1}^N \log x_i!$$

Step 3. Set the derivative to zero.

$$\ell'(\lambda) = -N + \frac{S}{\lambda} = 0 \implies \lambda = \frac{S}{N} = \bar{x}.$$

Step 4. Make sure that the solution is a maximum.

If $S > 0$, then $\ell''(\lambda) = -S/\lambda^2 < 0$ for all $\lambda > 0$. Thus ℓ is strictly concave, and the solution of Step 3 is the global maximum. If $S = 0$, then all $x_i = 0$ and $\ell(\lambda) = -N\lambda$. This function decreases, so the maximum is at $\lambda = 0 = \bar{x}$.

The maximum likelihood estimate is the sample mean: $\hat{\lambda}_{\text{ML}} = \bar{x}$.

5. Let $x_1, x_2, \dots, x_N$ be an independent sample from $\text{Poiss}(\lambda)$. The prior distribution is the gamma distribution

$$p(\lambda) = \mathcal{G}(\lambda \mid a, b) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda), \quad a, b > 0.$$

Find the Bayesian estimate of λ as the mean of the posterior distribution $p(\lambda \mid x_1, \dots, x_N)$.

Idea. Bayes' theorem gives the posterior distribution:

$$p(\lambda \mid X) = \frac{p(X \mid \lambda) p(\lambda)}{\int p(X \mid \lambda) p(\lambda) d\lambda}, \quad X = (x_1, \dots, x_N).$$

The numerator and the denominator contain the same function. The denominator only integrates it over λ . Thus you do not have to calculate the integral directly:

- (a) Write the numerator as $C \cdot g(\lambda)$. The constant C does not depend on λ .
- (b) Find a known distribution q with a density proportional to g : $g(\lambda) = Z q(\lambda)$, where Z is a known constant.
- (c) A density integrates to 1. Thus the denominator is $\int C Z q(\lambda) d\lambda = C Z$.
- (d) Divide: the posterior distribution is q .

The same method gives the posterior mean without a direct integral.

Solution. Let $S = \sum_{i=1}^N x_i$.

Step 1. Write the numerator as a constant times a function of λ .

$$p(X \mid \lambda) p(\lambda) = \prod_{i=1}^N e^{-\lambda} \frac{\lambda^{x_i}}{x_i!} \cdot \frac{b^a}{\Gamma(a)} \lambda^{a-1} e^{-b\lambda} = \underbrace{\frac{b^a}{\Gamma(a) \prod_i x_i!}}_C \cdot \underbrace{\lambda^{a+S-1} e^{-(b+N)\lambda}}_{g(\lambda)}.$$

Step 2. Find the distribution with a density proportional to g .

Let $a' = a + S$ and $b' = b + N$. The function g has the form of the gamma density with these parameters:

$$g(\lambda) = \lambda^{a'-1} e^{-b'\lambda} = \frac{\Gamma(a')}{b'^{a'}} \mathcal{G}(\lambda \mid a', b').$$

Step 3. Calculate the denominator.

The gamma density integrates to 1:

$$\int C g(\lambda) d\lambda = C \frac{\Gamma(a')}{b'^{a'}} \int \mathcal{G}(\lambda \mid a', b') d\lambda = C \frac{\Gamma(a')}{b'^{a'}}.$$

Step 4. Divide the numerator by the denominator.

The constants cancel:

$$p(\lambda \mid X) = \mathcal{G}(\lambda \mid a + S, b + N).$$

Step 5. Calculate the posterior mean.

The function $\lambda^{a'} e^{-b'\lambda}$ has the form of the density $\mathcal{G}(\lambda \mid a' + 1, b')$. Thus its integral is $\Gamma(a' + 1)/b'^{a'+1}$, and

$$\mathbb{E}[\lambda \mid X] = \int \lambda \mathcal{G}(\lambda \mid a', b') d\lambda = \frac{b'^{a'}}{\Gamma(a')} \int \lambda^{a'} e^{-b'\lambda} d\lambda = \frac{b'^{a'}}{\Gamma(a')} \cdot \frac{\Gamma(a' + 1)}{b'^{a'+1}} = \frac{a'}{b'}.$$

The last equality uses $\Gamma(a' + 1) = a' \Gamma(a')$. The Bayesian estimate is

$$\hat{\lambda}_B = \frac{a + S}{b + N}.$$

Conclusion.

- The gamma prior and the Poisson likelihood give a gamma posterior. A prior with this property is *conjugate* to the likelihood. With a conjugate prior, the update only changes the parameters: $a \rightarrow a + S$ and $b \rightarrow b + N$. Thus a acts as a prior number of events, and b acts as a prior number of observations.
- The estimate is a weighted mean of the prior mean a/b and the maximum likelihood estimate $\bar{x}$ from Problem 4:

$$\frac{a + S}{b + N} = \frac{b}{b + N} \cdot \frac{a}{b} + \frac{N}{b + N} \cdot \bar{x}.$$

When N increases, the weight of the data increases, and the estimate goes to $\bar{x}$.

- When $a \rightarrow 0$ and $b \rightarrow 0$, the prior density becomes proportional to $1/\lambda$. This is the non-informative prior from the notes below. With this prior, the Bayesian estimate is equal to $\bar{x}$.

Notes on comparing the estimates from Problems 4 and 5.

1. For more about non-informative priors, see Section 5.4 “Priors” in Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*.
2. The prior $p(\lambda) \propto \frac{1}{\lambda}$ is a uniform prior on $\log \lambda$. To see this, use the change-of-variables rule for the density of a function of a random variable. See, for example, [this article](#).